

DAVID (SEHEON) CHANG

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Education

University of Texas at Austin

Bachelor of Science in Computer Science

Bachelor of Science in Electrical and Computer Engineering

Aug 2023 – May 2027

Austin, Texas

University Honors

Experience

2nd Infantry Combined Division - U.S. 8th Army

Sergeant

Jun 2025 – Dec 2026

Pyeongtaek, South Korea

- Mandatory service as a 56M KATUSA Sergeant; collaborated with the United Nations Command (UNC) to coordinate historical training, provide tactical translation, and brief senior political and military leadership.
- Awarded the Army Commendation Medal (ARCOM) and Army Achievement Medal (AAM) by Major General Lombardo for exceptional translation and operational support during combined field training exercises.

Ajou University School of Medicine

Research Intern

Jun 2024 – Sep 2024

Suwon, South Korea

- Utilized SQL and R to map heterogeneous databases into OHDSI's Common Data Model (CDM) standards.
- Co-authored two clinical AI papers published in JMIR (JIF 6.0) utilizing BERT and GPT-based NLP pipelines.

LooxidLabs

Project Intern

Jun 2022 – Jan 2023

Seoul, South Korea

- Developed 'CarePlay,' a VR mental well-being service in Unity using EEG neurofeedback; supervised trials in NYC.

Projects

Tessera | *ESP32, C/C++, PN532 NFC, PySide6, AES-256-GCM, local SQLCipher*

Mar 2026 – Present

- Led a team to build an offline emergency medical records dog tag scanner; placed 7th of 700 teams in the ROK Army Startup Competition and awarded by the ROK Department of Defense.
- Engineered ESP32 firmware to read and decrypt compressed, AES-256-GCM encrypted records over SPI in under 1s, and built a PySide6/SQLCipher desktop application for offline serial provisioning.

Sisypus | *C#, .NET 8, WPF, Windows Registry, Windows Internals (IFEO)*

Apr 2026 – Present

- Developed a native Windows focus utility that intercepts game launches to inject a WPF todo checklist prior to launch, utilizing native Windows Registry Image File Execution Options (IFEO) to avoid CPU-heavy polling loops.

Hearty | *Next.js, Python, AWS (Chalice - Lambda, API Gateway / CloudFront / S3), Neon Postgres*

Apr 2026 – Present

- Created a serverless voice diagnosis platform utilizing a Next.js single-page application hosted on S3 and CloudFront, backed by Python AWS Chalice microservices and a serverless Neon PostgreSQL database.

Alpaca Trading Bot | *Python, SQLite (WAL Mode), Streamlit, systemd, Telegram API*

Jan 2026 – Present

- Engineered a 24/7 autonomous stock trading engine deployed on a Vultr VPS, managed via systemd unit files, utilizing SQLite Write-Ahead Logging (WAL) for telemetry tracking, a Streamlit dashboard, and Telegram alert webhooks.

Ajou Course Manager | *Django, Docker, React, PDFMiner, OpenAI API, PostgreSQL*

Oct 2025 – Present

- Built a course syllabus optimizer with a PDFMiner data-extraction pipeline, prompting GPT models to classify course plans and syllabus guidelines, and persisting metrics in a clean PostgreSQL instance.

Campus Involvement / Organizations

Faculty Applicant Interviewer

Chandra Family Department of Electrical and Computer Engineering

Spring 2024 – Present

University of Texas at Austin

- Selected by faculty and advisory board to participate in faculty hiring interviews for the ECE department.

Texas Convergent

IoT Build Team Tech Member

Spring 2024 – Present

University of Texas at Austin

- Identifying and approaching issues with hands-on IoT technology solutions. Currently at project planning stage.

Directed Reading Program

Natural Language Reasoning Team

Spring 2024 – Present

University of Texas at Austin

- Engaged in research discussions on CoT and o1 reasoning, advised by Ph.D. student Zayne Sprague.

Additional Information

Languages: ARM Assembly, C, C++, C#, Java, JavaScript, Python, TypeScript | **Tools/Frameworks:** .NET, AWS, Django, Docker, Git, Kubernetes, Linux, Next.js, Node.js, PostgreSQL, React, SQLite, Streamlit